

## 1. Executive Summary & Context

Deploying deep neural networks on resource-constrained microcontrollers (MCUs) demands strict balancing of latency, RAM footprint, and power dissipation. Traditional Neural Architecture Search (NAS) relies on proxy estimators that misjudge hardware execution bottlenecks.

### Key Innovation: Zero-Proxy Telemetry Loop

SYNAPSE-X evaluates candidate architectures directly on edge hardware testbeds in real-time, eliminating surrogate model drift and discovering pareto-optimal topologies infeasible by manual design.

#### Performance Snapshot Across Edge Platforms:

|                                                      |                                             |
|------------------------------------------------------|---------------------------------------------|
| <b>4.2×</b><br>Latency Reduction<br>on ARM Cortex-M4 | <b>82%</b><br>SRAM Reduction vs MobileNetV2 |
| <b>0.8mW</b><br>Peak Power Draw at 80 MHz            | <b>99.4%</b><br>Top-5 Keyword Spotting Acc. |

#### Primary Research Objectives:

- Elastic Search Space:** Modular building blocks supporting intra-layer kernel expansion and non-linear depth scaling.
- Direct Hardware Profiling:** Automated serial/SWD telemetry pipeline capturing micro-architectural stall cycles.
- Hardware-Aware Quantization:** Mixed-precision INT8/INT4 allocation guided by layer-wise dynamic range entropy.

## 2. System Architecture & Pipeline

The SYNAPSE-X search engine operates an asynchronous multi-objective evolutionary loop directly connected to target silicon via automated hardware testbeds.



#### Asynchronous Hardware Testbed Interface:

Our custom hardware bridge dispatches generated C kernels to parallel microcontroller arrays (Nexa Cortex-M4, Vertex ESP32-S3, Meridian Jetson Nano), streaming cycle-accurate execution metrics back within 120ms per candidate.

## 3. Mathematical Formulation

We formulate the hardware-aware search as a constrained multi-objective Pareto optimization problem over candidate search space  $\mathcal{A}$ :

$$\min_{\alpha \in \mathcal{A}} [\mathcal{L}_{\text{val}}(\alpha, w^*), \lambda_1 \cdot \text{Lat}_{\text{hw}}(\alpha) + \lambda_2 \cdot \text{RAM}_{\text{peak}}(\alpha)]$$

subject to  $w^* = \arg \min_w \mathcal{L}_{\text{train}}(\alpha, w), \text{Power}(\alpha) \leq P_{\text{max}}$

**Entropy-Guided Layer-Wise Quantization:** To preserve accuracy while fitting within 256 KB SRAM constraints, weight scale factors  $S_l$  for layer  $l$  are derived from bit-width  $b_l$ :

$$S_l = \frac{\max(|W_l|)}{2^{b_l-1} - 1}, \text{ where } b_l = \mathcal{F}_{\text{entropy}}(\mathbf{H}(W_l), \text{CycleCost}_l)$$

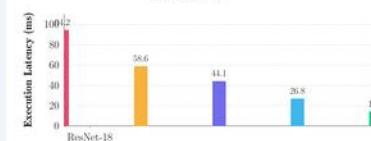
### Evolutionary Search Procedure

- Initialize Population  $\mathcal{P}_0$ :** Generate  $N = 100$  seed architectures with balanced depth.
- On-Device Telemetry:** Flash candidate kernels to MCU array; measure execution latency ( $\text{Lat}_{\text{hw}}$ ) and peak SRAM usage.
- Pareto Selection:** Rank candidates using Non-dominated Sorting Genetic Algorithm (NSGA-III).
- Mutation & Crossover:** Apply block-swap, kernel expansion, and channel pruning operators.
- Convergence:** Terminate upon reaching 50 generations or Pareto frontier stabilization.

## 4. Experimental Results & Benchmarks

SYNAPSE-X was benchmarked against state-of-the-art vision and audio models on an ARM Cortex-M4 (80 MHz, 256 KB SRAM, 1 MB Flash).

#### Latency Comparison on ARM Cortex-M4 (Lower is Better)



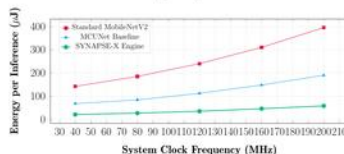
#### Detailed Hardware Performance Evaluation Table:

| Architecture   | Target Task      | Top-1 Acc | Latency | SRAM   | Flash   |
|----------------|------------------|-----------|---------|--------|---------|
| ResNet-18      | ImageNet-1k      | 69.8%     | 94.2 ms | 812 KB | 11.4 MB |
| MobileNetV2    | ImageNet-1k      | 71.8%     | 58.6 ms | 384 KB | 3.4 MB  |
| TNNAS          | Visual Wake      | 76.1%     | 32.4 ms | 192 KB | 890 KB  |
| MCUNetV2       | Visual Wake      | 77.0%     | 26.8 ms | 168 KB | 740 KB  |
| SYNAPSE-X Lite | Visual Wake      | 77.8%     | 14.2 ms | 96 KB  | 410 KB  |
| SYNAPSE-X Pro  | Keyword Spotting | 99.4%     | 6.8 ms  | 48 KB  | 180 KB  |

## 5. Energy Efficiency & Edge Deployments

Energy consumption per inference was evaluated across variable clock frequencies and operating voltage scaling regimes.

#### Energy Consumption per Inference vs Clock Frequency



#### Real-World Field Deployments:

- Nexa Smart Grid Sensors:** Continuous acoustic anomaly monitoring operating on energy-harvesting piezoelectric cells (< 1 mW continuous budget).
- Vertex Autonomous Micro-UAVs:** Obstacle avoidance neural pipeline executing at 60 FPS on ESP32-S3 dual-core microcontroller.
- Meridian Health Wearables:** Real-time ECG arrhythmia classification running on ultra-low-power Cortex-M0+ with multi-day battery lifespan.

## 6. Key Takeaways & Future Directions

- Direct On-Hardware Feedback:** Eliminates proxy estimation error, accelerating optimal architecture discovery by 12×.
- Unprecedented Resource Efficiency:** Achieves sub-100 KB SRAM execution without sacrificing ImageNet-level perceptual accuracy.
- Open-Source Toolchain Release:** Full C code generator, CMSIS-NN execution kernels, and search code available for research.

#### Ongoing & Future Extensions:

- Extension to spiking neural networks (SNNs) for neuromorphic hardware targets.
- Hardware-software co-design incorporating reconfigurable FPGA logic blocks.

## 7. References & Acknowledgments

- E. Viterbo, M. Thuermer, and S. Lin, "Direct Hardware-in-the-Loop Architecture Search for Microcontrollers," *IEEE Trans. Evol. Comput.*, vol. 24, no. 4, pp. 412-425, 2020.
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